# WindBrowser manual

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# 1 WindBrowser

*This manual describes WindBrowser 1.0.176-beta.*

*Suomenkielinen ohje (PDF, Finnish manual): https://pclusius.github.io/windbrowser-docs/WindBrowser-manual-fi.pdf*

WindBrowser is an Android app that downloads wind forecasts from the Finnish Meteorological Institute (FMI) and, with your own free key, Météo-France for map areas you choose, and **keeps every forecast run** on the phone. It is meant for judging how far a forecast can be trusted, for example before a sail race or a cruise leg on the Baltic Sea.

## 1.1 Why keep every run?

FMI serves only the latest run of each model. When a new run replaces the old one you cannot see how the forecast has changed. WindBrowser stores each run it downloads, so you can compare successive runs for the same place and time. If they agree, the forecast is probably reliable; if they differ a lot, it is not.

History exists only from the moment the app starts downloading, so add your area a few days before you need it.

## 1.2 Models

| Model | Resolution | Area size limit | Data per square degree and run |
|---|---|---|---|
| **HARMONIE** | fine (about 2.5 km) | 16 square degrees | about 385 kB |
| **ARPEGE** (Météo-France) | medium (0.1 degree, about 11 km) | 400 square degrees | about 35 kB |
| **ECMWF-IFS** | coarse (0.25 degree) | 1600 square degrees | about 2 kB |

HARMONIE covers northern Europe. ARPEGE covers Europe and the Baltic and needs your own Météo-France key (ARPEGE). ECMWF-IFS works anywhere in the world. They can be combined: ECMWF-IFS under HARMONIE gives a wide overview with detail where you need it, and a third model shows how far the models agree.

### 1.3 How to read this manual

The chapters follow the order in which you will use the app:

1. First start: choose your first area.
2. The map: the wind layer and controls.
3. Charts: tap a point and compare runs.
4. Tiles: manage your areas.
5. ARPEGE and the key: Météo-France's model and its key.
6. Stations: observed wind.
7. Routing: plan a passage on your own route, or let the app find the best route and departure.
8. Settings
9. Updates and troubleshooting
10. About and credits

WindBrowser needs Android 10 or newer and works on phones and tablets.

!!! note The app is for use over the sea. Observation stations are a curated set of marine stations.

## 2 First start

When you open WindBrowser for the first time the map is empty. **Nothing is downloaded until you have chosen an area.**

The menu (hamburger button, top left) holds **Add new forecast tile** at the top:

### 2.1 Add an area

1. Open the menu with the **hamburger button** at the top left of the map.
2. Pick **Add new forecast tile**.
3. Choose the **model**: HARMONIE (detailed, small areas), ARPEGE (medium, Europe and the Baltic, needs your own free Météo-France key, see ARPEGE) or ECMWF-IFS (coarse, large areas, very small downloads).
4. A rectangle appears inside the current view. Drag its **top-left and bottom-right handles** to cover the sea area you want. Touches elsewhere pan the map.
5. The dashed outline around your rectangle is the area that will really be downloaded. It is slightly larger so that points at the edge of your rectangle can still be interpolated.
6. For ARPEGE, choose **how many days ahead** it is downloaded with the chips (1–4, default 3); see ARPEGE. Check the **disk estimate** in the panel. A warning appears if the data limit would be exceeded.
7. Confirm. The first forecast starts downloading ("Downloading the first forecast…"). When data is ready, the panel under the map says "Tap the map, or an orange station").
8. The app now asks once whether it may **run without battery restrictions** ("Keep the forecasts coming?"). Allow it if you want no gaps in the forecast history; see Updates. You can change it later in Android's settings: App info → Battery → Unrestricted.

!!! tip For a sail race or just a cruise leg, choose a HARMONIE area around the course and a larger ECMWF-IFS area around it. Both can be active at once.

### 2.2 Size limits

An area can hold at most 16 square degrees with HARMONIE, 400 with ARPEGE and 1600 with ECMWF-IFS. Larger rectangles are refused. See Tiles for what happens when areas overlap.

### 2.3 What happens next

The app checks for a new forecast every hour in the background, and also when you bring it to the front (at most every 10 minutes). See Updates.

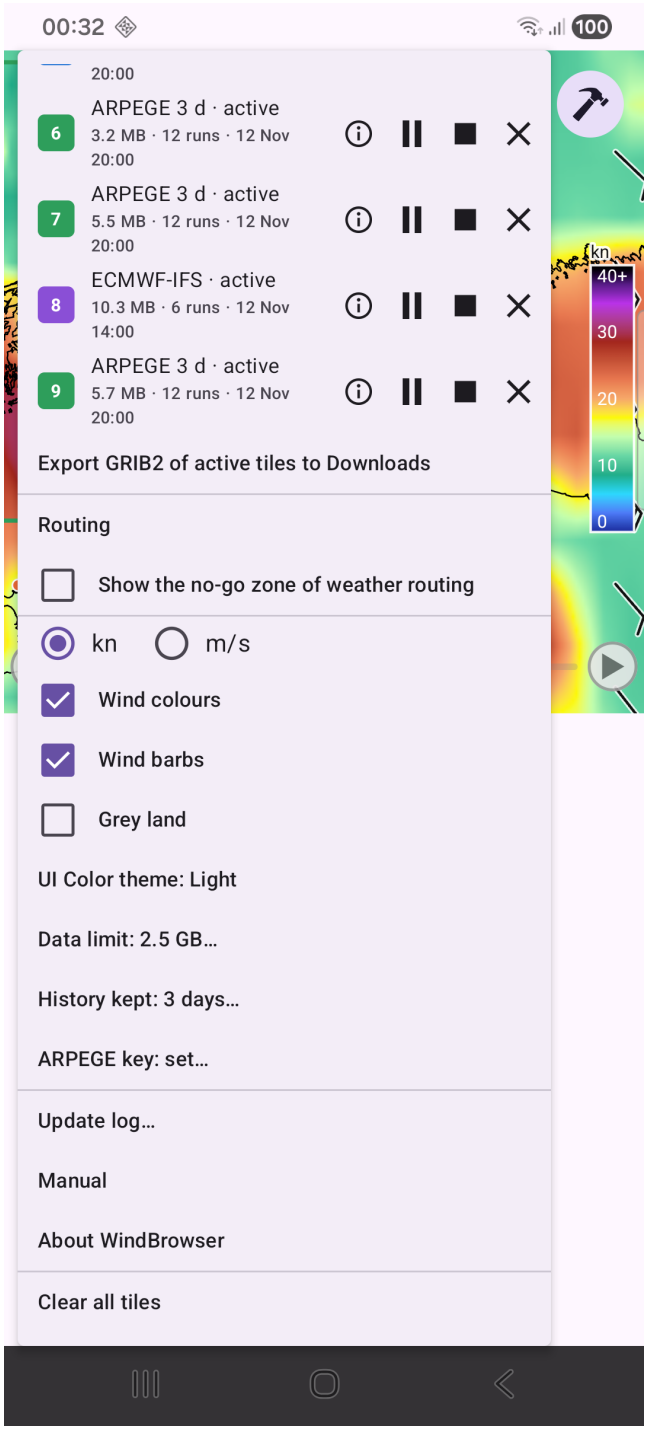



Figure 1: Menu

# 3 The map

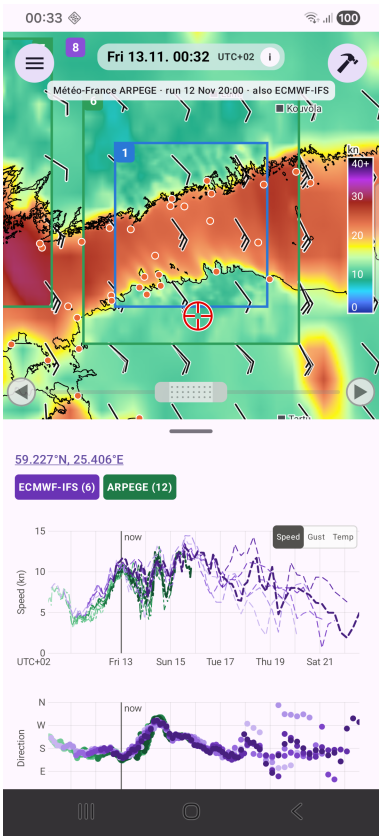
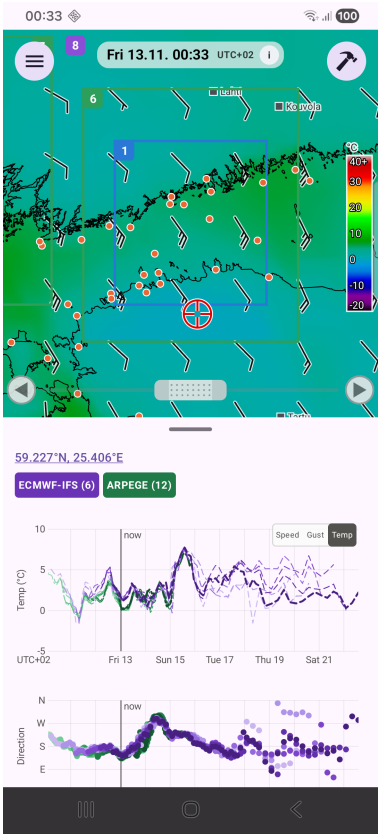



Figure 2: Map overview

## 3.1 What you see

- **Wind layer**: colours show wind speed, from blue (calm) through green, yellow, orange and red to purple and black (40 knots and more). Wind barbs show direction and speed in knots. The layer is opaque.
- **Colour bar** with the numbers inside it.
- **Temperature map**: choose **Temp** on the toggle in the speed chart and the colours show the 2 m air temperature from -20 °C (black) through purple, blue, cyan, green, yellow and orange to red and light grey at 40 °C and above (the `nipy_spectral` scale), with its own colour bar in °C. The wind barbs stay. ARPEGE and ECMWF-IFS have temperature, HARMONIE does not: the colours come from ARPEGE where it covers the point and otherwise from ECMWF-IFS, and the barbs from the topmost model, so where a HARMONIE tile lies over one of them the colours come from that one and the barbs from HARMONIE; a HARMONIE tile with nothing under it shows barbs only. The grey land setting is kept as it is, so with grey land on, the temperature shows over the sea only. Choose **Speed** to go back to the wind layer.




- **Coastline**: grey land with a black shoreline, drawn from a bundled vector map, so no map tiles are needed and the map works offline. - **Place names**: small names of larger towns; more appear as you zoom in. - **Station dots**: orange dots are observation stations (Stations); small when zoomed out, larger when zoomed in. - **Tile outlines**: the borders of your areas, with a number badge. Solid means active, dashed means paused or stopped. Blue is HARMONIE, green is ARPEGE, purple is ECMWF-IFS.

Zoom runs from level 5 to 15. Drag to pan, pinch to zoom.

## 3.2 The caption

The caption (shown with the small (i) button in the time box at the top of the map) reads, below that box„ for example, `ECMWF-IFS · Sat 7 20:00 UTC+02 · run 4 Nov 02:00`: the model, the forecast time shown and the model run it comes from. In the temperature map it also says “2 m temperature” and, where the barbs come from another model, “wind barbs FMI-HARMONIE”. For a tapped point it names the model that covers the point, and adds “also ECMWF-IFS” when a second model reaches it too. Two taps on the time box or on the scrubber’s handle put the map and the chart cursor at the actual time.

## 3.3 Old data fades

A tile whose newest run is more than 24 hours old fades to grey, fully grey at 48 hours. Grey means the data is stale, usually because downloads are paused or failing.

## 3.4 Two models together

The finest model is drawn on top: ECMWF-IFS below ARPEGE below HARMONIE, so you see the fine model where it exists and the coarser ones around it. To look at a layer that is covered, hide the ones above it by tapping their names in the chart legend (Charts).

## 3.5 Tools button

The round hammer button at the top right of the map holds three tools. Tap it and the radar, routing and location buttons fan out round it; tap one to use it. Tap the hammer again, or anywhere else on the map, to fold

them away. The menu button at the top left stays where it is.

- **Rain radar**: switches the rain radar layer on and off (see below).
- **Routing** (a zigzag route with a dot at each end): opens the same window as **Routing** in the menu: your routes, saved plans and "Plan a passage" (Routing).
- **Location**: finds your position (one fix; GPS if you allowed precise location), selects that point as if you had tapped it, and centres the map there. The position is not stored or sent anywhere.

### 3.6 Rain radar

The radar button is blue while the layer is on and grey while it is off. Switching it on shows the FMI rain radar (Finland and its surroundings) for the time on the clock, and moves the time to now. While the radar is shown it replaces the wind or temperature layer and its colour bar, and the time cannot be moved past now; switch the radar off to get the forecast back. The pictures of the last 6 hours are downloaded when you switch it on (one every 15 minutes), so scrubbing the time needs no network. Colours go from blue to dark red as the rain gets heavier (0.1 mm/h and up to 5 mm/h and more); a greyer tone with a thin line marks where the radars do not reach. Where there is no rain the picture is the sea colour: pale in the light theme and mid-grey in the dark one.

### 3.7 The time box and the scrubber

The time of the map and the charts, which are always the same, is shown in a faintly see-through box at the top of the map: the day and time large and the UTC offset small beside it. The box turns yellow when the time is more than a minute from now. **Double tap** it to go to now. The small (i) button inside the box shows the caption of the map (below the box); the caption hides again when you tap the map or the charts.

The scrubber at the bottom of the map has an arrow button near each edge of the screen, a faint line between them and a handle in the middle. The box is at the top so that your finger does not hide the time while you scrub.

- The arrow at each end runs the time forward or back by itself (a quarter of an hour every quarter second); the running arrow becomes a pause sign. Press the other arrow to turn the direction round.
- **Pull the handle** to the right or left to change the time: it runs forward or back for as long as you hold the handle away from the middle, and faster the further you pull (up to 60 dp; the speed ranges from 20 minutes of forecast per second to four times that). The handle springs back to the middle when you let go. The buttons and the line stay where they are.
- **Double tap** the handle to go to now.

### 3.8 Divider

Between map and charts there is a drag handle. Portrait: map above, charts below. Landscape: map on the left, charts on the right. Each orientation remembers its own split. On a tablet, or in any window at least 840 dp wide (also an unfolded foldable or a split screen), the map and the charts are side by side, as in landscape.

## 4 Charts

**Tap anywhere on the map** to select a point (a red crosshair marks it), or tap an orange station dot; the title then shows the station name or the coordinates. The charts show the forecast at that point from **every stored run** of every area that covers it, whether the tile is active or not.

### 4.1 Reading the charts

- **Speed chart** (top): each run is one line. HARMONIE is blue and solid; ARPEGE is green with long and short dashes; ECMWF-IFS is purple and dashed. Older runs are paler, the newest is darkest. With only one model the legend reads "Forecast runs, old → new".
- **Direction chart**: shown as dots, never lines, because 359 and 1 degrees are neighbours.
- **Speed / Gust / Temp toggle**: small switch in the corner of the speed chart. Gust exists for HARMONIE and ARPEGE and temperature (2 m air temperature, in °C) for ECMWF-IFS and ARPEGE; each option

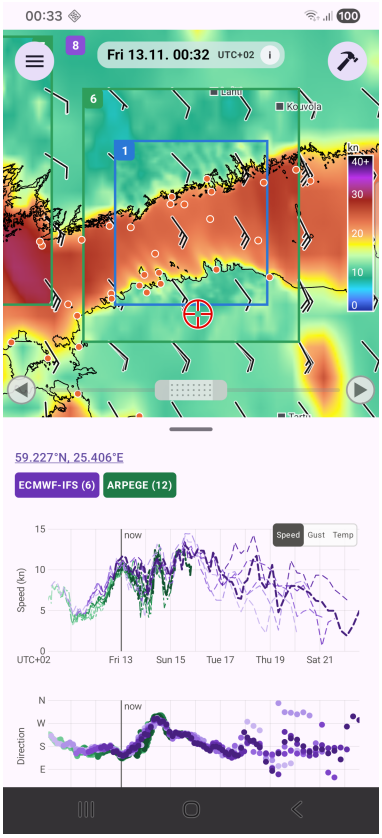



Figure 3: Charts

is greyed out where no run has it. Choosing **Temp** also turns the map's colours to temperature (see The map). If you tap a point that no temperature forecast covers, the chart and the map go back to speed.
- **Observed wind** from a tapped station is drawn in orange. With **Temp** chosen it is the observed air temperature (FMI, SMHI, DMI and Marine Helsinki stations).

If the runs lie close together the forecast is consistent. A wide fan means uncertainty.

## 4.2 The cursor and the readout

Touch a chart with **one finger** to move the cursor. The map follows the cursor in time. The line above each chart reads, for example:

```
HARMONIE: 12.2 (11.1–13.0)  ARPEGE: 13.4 (12.0–14.1)  ECMWF-IFS: 15.2 (11.1–15.2)  Observed: 13.2
```

Before you touch the chart the models show a dash (–).

For each model it shows the value of the newest run covering that time, and in brackets the lowest and highest of all runs covering it, then the observation.

## 4.3 Legend

The legend shows each model's name in bold and in the colour of its lines, with its number of runs, and "Observed" in orange. **Tap a name** to hide that model: its lines, dots and readout leave the charts, and its **map layer, wind barbs and caption leave the map**, which is the way to see a layer that lies under another one. Tap again to show it. The names turn grey while a model is hidden. Choosing another point shows everything again.

## 4.4 Zoom

**Two fingers** zoom and pan the shared time axis. Speed rescales to what is in view and hourly ticks appear when zoomed in. **Tap for now** resets the axis.

## 4.5 Auto-advance

The two arrows at the ends of the scrubber at the bottom of the map move the cursor, and so the map, forward or back by themselves: a quarter of an hour per quarter second, from the cursor (or from now) to the end of the stored forecasts. Pulling the scrubber's handle, the arrow again (it shows a pause sign while running) or two taps on the time box stop it (The map).

## 4.6 Sun and moon

The place in the title above the charts, the coordinates or a station's name, is underlined: **tap it** to see the sun and moon for that place.

- The pop-up opens on the day of the chart cursor, or on today. The left and right arrows step one day at a time, and **Today** returns to today.
- The **day strip** runs from midnight to midnight and is coloured by how far the sun is below the horizon: night, astronomical, nautical and civil twilight, and day. Red marks show sunrise and sunset, the dot is solar noon, and the bar under the strip shows when the moon is up, with its rise and set marked in red.
- The **table** gives the length of daylight (sunrise to sunset) and of the civil, nautical and astronomical day: the time the sun is more than 6, 12 or 18 degrees below the horizon is the twilight, so the dark time of each kind is the rest of the 24 hours. Where the sun does not cross a level, as on a Baltic summer night, the table says "all day" or "never".
- The **moon** is drawn as it looks that day (flipped in the southern hemisphere), with the phase, the share lit, its age in days, moonrise and moonset, and the next new and full moon.

Everything is calculated from the position and the date, so it works without any forecast or network. The sun is accurate to about 2 minutes and the moon to about 5. The times are on the phone's clock and its UTC offset, and a "day" is the phone's calendar day.

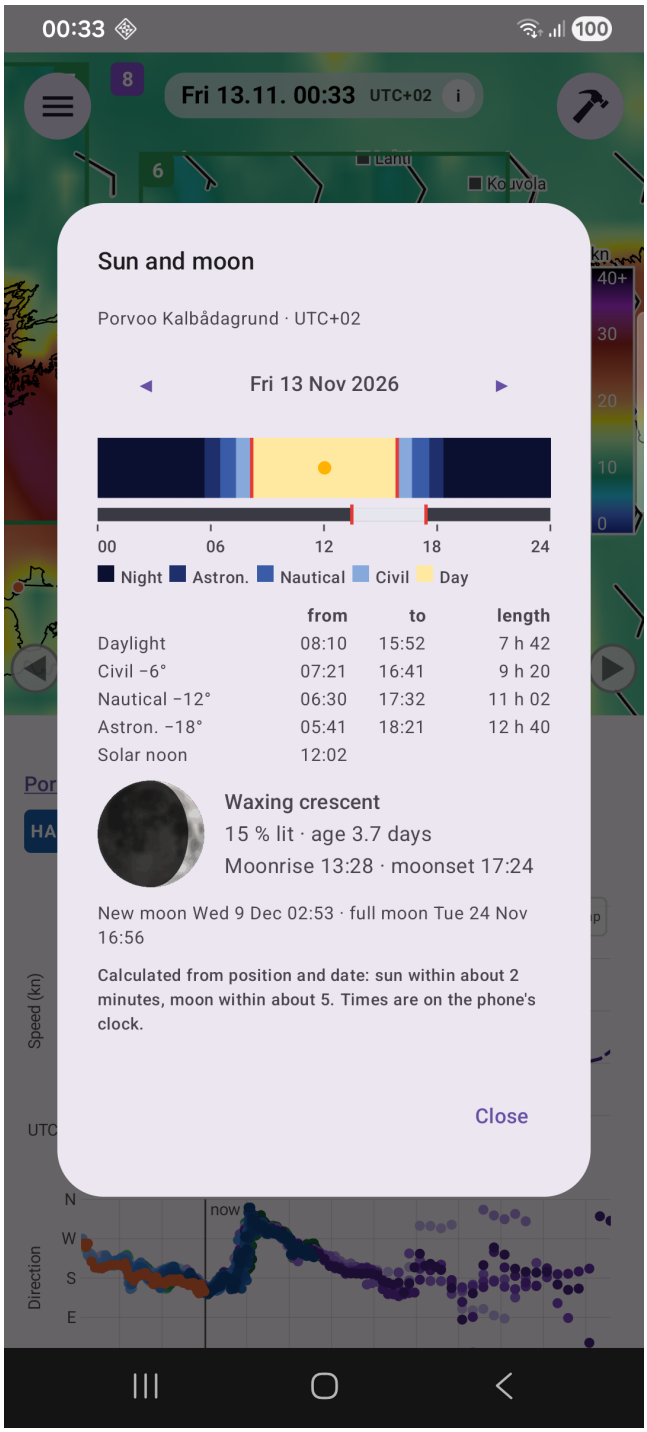



Figure 4: Sun and moon

### 4.7 Units

Speed is shown in knots or m/s (Settings). Times show the phone's UTC offset, for example `UTC+02`.

## 5 Tiles

An area you download is called a **tile**. You can have as many as you like, numbered from 1 (the data limit is what bounds them). Everything is managed from the hamburger menu.

### 5.1 The tile list

Each row shows the tile's number (blue badge HARMONIE, green ARPEGE, purple ECMWF-IFS), model, state, size, number of runs and newest run, for example `HARMONIE · active · 4.9 MB · 9 runs · 4 Nov 08:00`. An ARPEGE tile also shows its forecast length, for example `ARPEGE 3 d · active`. Tap a row to zoom the map to that tile.

| Button | Action |
|---|---|
| **i** | List the stored runs (model run time and file size) |
| **pause / play** | Pause stops downloads but keeps the data; play resumes. Takes effect at once, also during a download |
| **stop** | Stops downloads; the tile is removed when its data has expired |
| **delete** | Removes the tile and its data at once, even if it is downloading. Always asks for confirmation |

States: **Active** downloads; **Paused** keeps data but does not download; **Stopped** is removed once its data has expired.

### 5.2 Overlap

Only active tiles of the same model count.

- An area completely covered by existing tiles is refused.
- If 70 % or more is covered by one tile, the app offers to **enlarge that tile** to hold both, or to add the area anyway.
- A smaller overlap is accepted with a note that the overlap is downloaded twice.

### 5.3 Storage limits

- **Data limit**: 1 to 5 GB (default 2). When exceeded, the oldest runs go first, those of non-active tiles before active ones; the newest run of an active tile is never removed. A dialog tells you once what was removed and what you can do: shorter history, smaller areas or fewer tiles, ECMWF-IFS, or a higher limit.
- **History**: 1 to 14 days, set with a slider in **History kept…**. Shortening it deletes the older runs when you close the dialog.

**Clear all tiles** always asks for confirmation.

## 6 ARPEGE and the Météo-France key

**ARPEGE** is the global forecast model of Météo-France. WindBrowser uses its 0.1 degree (about 11 km) version over Europe and the Baltic, with wind, **gusts** and **2 m air temperature**. It gives four runs a day (00, 06, 12 and 18 UTC) and forecasts up to 102 hours ahead.

Unlike FMI's models, Météo-France gives its data only to users who have **their own free key**. The key is kept on your phone only and is sent only to Météo-France. Without one, ARPEGE cannot be chosen when you add an area, and an ARPEGE tile that is already there says it needs a key.

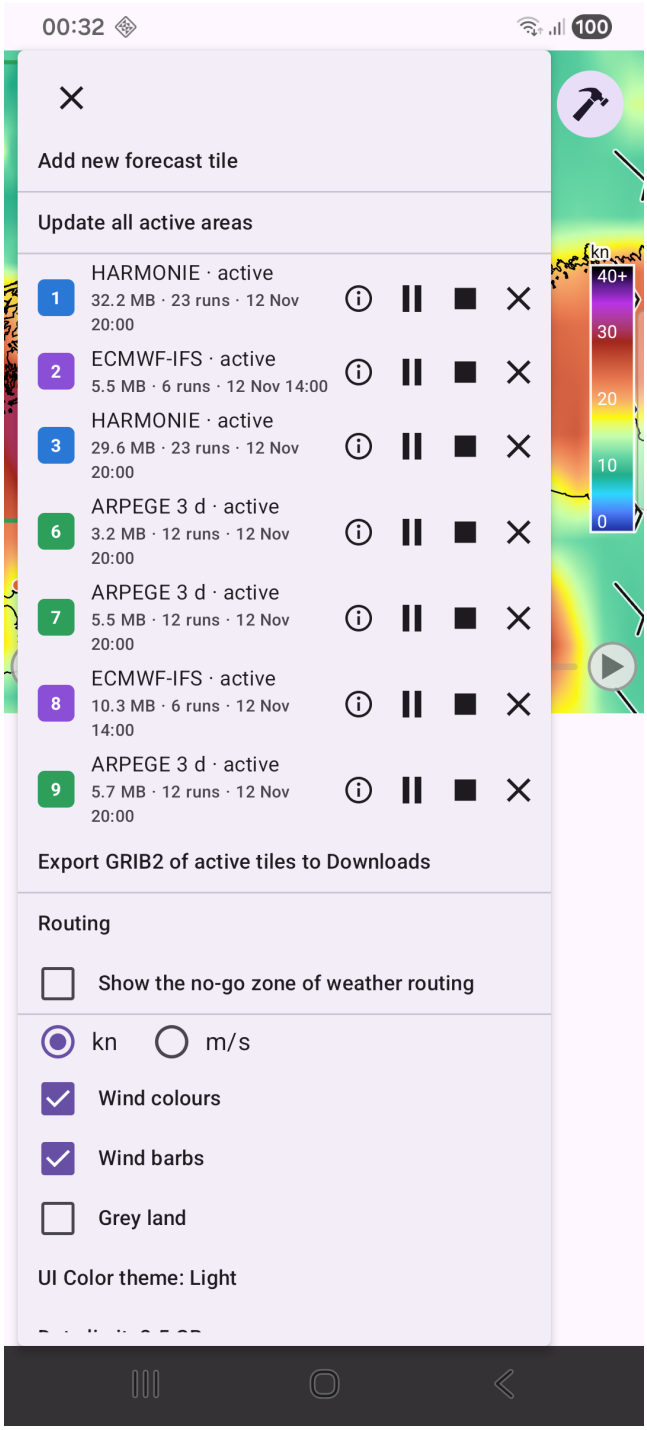



Figure 5: Tile list

### 6.1 Get a key

1. Make a free account at **portail-api.meteofrance.fr**.
2. Subscribe to the API called **API Modèle ARPEGE**.
3. On the API's page, under *Configure API*, choose the token type **API Key**, set how long it should last (365 days works) and press **Generate Token**.
4. **Copy all of the key.** It is one very long line of text, about 1,500 characters, with no spaces. Click in the key field, use *select all* and copy; a double click selects only a part of it.

!!! note Making a new key on the portal may stop the old one from working. If downloads stop with "The Météo-France API key was rejected", the key has expired or been replaced: make a new one and enter it again.

### 6.2 Enter the key in the app

Menu → **ARPEGE key: not set...**, paste the key and press **Save**. The dialog shows how many characters it got and refuses a key that is clearly too short, which usually means it was cut when copied. After saving, the menu says **ARPEGE key: set...**; open it again to replace the key, or press **Remove** to delete it. If an ARPEGE tile is waiting, an update starts at once.

### 6.3 Add an ARPEGE area

Choose **ARPEGE** (green) when you add an area (First start). An area can hold at most 400 square degrees, and the data is small, about 35 kB per square degree and run for three days of forecast. ARPEGE covers 20°N to 72°N and 32°W to 42°E.

### 6.4 How far ahead

When you add an ARPEGE area, the area editor has chips for how far ahead that tile is downloaded: 1 to 4 days, 3 by default. Each ARPEGE tile has its own length, shown in the tile list (`ARPEGE 3 d`) and used in the size estimate; to change it, add the area again with another length. Météo-France sends one small file for every variable and hour, so a longer forecast means more requests (3 days is about 290) and a longer update. The portal allows 100 requests a minute and the app spaces them out, so the first download of a run takes about **three minutes**.

### 6.5 What to expect

ARPEGE is drawn with green lines in the charts, with long and short dashes, and its tile outline and map layer are green. It is coarser than HARMONIE and finer than ECMWF-IFS, so on the map it lies between them. Gust and temperature work for ARPEGE as for the other models (Charts).

## 7 Routing: plan a passage

The routing part of WindBrowser answers: *if I sail this route, leaving at this time, when do I arrive, how fast do I go, and what wind do I meet?* The route is yours, from your chart plotter or Navionics, and the app sails it on the computer with the forecasts stored on the phone. By default it follows your line exactly. If you tick **Find the best route between the route's points (weather routing)**, it looks for the fastest line (or the one that best avoids what you want to avoid) between your points, using the forecast, the polar and the tacking angles.

Open it from the map menu: **Routing**, or with the routing button in the tools (hammer) button at the top right of the map.

### 7.1 Import a route

**Import route (GPX)...** reads a GPX file with a route (`rte`), a track (`trk`) or waypoints (`wpt`), up to 5 MB and 2,000 points. Navionics routes work as they are. Imported routes are listed with a checkbox (show or hide on the map), a coloured dot, and **Remove**. Tapping a route zooms the map to it.

**Route points.** Tap a point of a visible route on the map to open its window: the point's number, how weather routing passes it (**Through**, **Leave to port** or **To starboard**; the first and last point are always passed through), and its latitude and longitude, which you can type. **Save** stores the change, **Delete point** removes the point after asking (this cannot be undone), **Cancel** closes the window. A point that is left to port or starboard is drawn with an arrow circling it: clockwise for a mark left to starboard, counter-clockwise for one left to port. The choice matters only to weather routing; a route sailed as drawn goes through every point.

## 7.2 Boat and routing settings

The boat and routing settings are the lower part of the **Plan a passage…** dialog, below the forecast choices. They belong to the plan: a new plan starts with the settings of the last one, and each saved plan keeps its own, so **Make again with the latest run** and saved plans use the settings they were made with. Nothing about a boat is built into the app: import your own polar.

- **Polar.** Choose the polar in the drop-down. **Import polar (.pol)…** (the last item of the drop-down) reads an Expedition-style table: wind angles in rows, wind speeds in columns, boat speed in knots, separated by tabs or semicolons. You can import several polars and pick one for each plan. **Delete** beside the drop-down removes the chosen polar from the phone after asking; plans made with it keep their results, but to make one of them again you must import the polar again.
- **Upwind.** The boat tacks when the course is closer to the wind than the tack angle. Tick **Optimum tack angle from the polar** to use the best angle for each wind speed, or give a **Minimum tack angle**. With both, the larger is used. **Leeway when beating** (default 4°) makes the track lie further from the wind than the heading; it falls to 2° at 90°, 1° at 130° and 0° at 180°. **Progress when tacking (%)** is the share of the speed along the course that is kept while the boat tacks (default 95 %, blank is 100 %); the loss grows from nothing at the edge of the tacking angle to the full value 3° inside it.
- **Downwind.** The same in reverse: **Optimum downwind angle** (this assumes the boat gybes) and/or a **Maximum downwind angle**. With both, the smaller is used. A maximum like 178° means no gybing until the course is deeper than that, which is slower. **Progress when gybing (%)** works like the tacking one.
- **Engine.** Tick **Motor when the sailing speed is low** and give the motoring speed and the sailing speed under which the engine starts. It stops again when the sailing speed is 0.5 kn above that. Without an engine the boat waits for wind.
- **Performance. General boat speed** as a percentage of the polar (blank is 100 %, 90 is ten percent slower), and speeds as a percentage of the polar in light wind (below a wind speed you give), in strong wind (above one) and at nautical night (sun more than 12° below the horizon). The adjustments multiply: they come on top of the general one.
- **Avoid** (for the best route and departure). What the weather routing and *Best departure* below try to avoid; see those sections. The underlined **Parameters** link beside the heading opens the weights and the integration step.
- **When the forecast ends.** Go on with the last wind, or stop the plan there.

The routing, too, uses only the mean wind, not gusts. Where the model forecasts gusts (HARMONIE and ARPEGE, not ECMWF-IFS) they are kept with the plan and shown on the wind graph and in the summary.

## 7.3 Plan a passage

**Plan a passage…** asks for the route (with **Reverse route** to sail it from the last point to the first, swapping marks left to port and starboard), the weather-routing tick, the departure time (this phone's time) and the forecast model, and holds the boat and routing settings above. **Cancel** takes you back to the **Routing** dialog. The newest run of that model is used; where it has no forecast (outside your tiles, or after the end of its run) the next less detailed model fills in. The plan is worked out in steps of five minutes along the route.

## 7.4 Weather routing

With **Find the best route between the route's points (weather routing)** ticked, the app searches for the best line along your route. Your points are kept and each is passed as set for it. The route is cut at the points that are passed **Through**, and each part is searched in the way that suits it: a part with points left to port or

starboard is searched on a grid by Dijkstra's algorithm, where a point only has to be left on its side and the line does not have to go near it when it need not; a part without such points is searched by the isochrone method (see About for the reference), which gives smoother lines. The cost of a line is its trip time plus, for what you ticked under **Avoid**, the hours that meet each condition times its weight, so with nothing ticked it is simply the fastest line. The search uses the forecast, the polar, the tacking and gybing angles and losses, the engine rule and the other settings of the plan.

- **While it runs**, the map shows the best partial line and, for an isochrone search, the isochrones (the fronts the search has reached) and the stage; for a grid search the grid points that are settled twinkle on the map, spreading from the start, and a bar in the panel under the map shows roughly how far the search is (an estimate that compares the time reached with the time the part takes sailed as drawn). The panel also tells which part of the route is being searched. A search takes from a few seconds to a minute, depending on the route and the phone. **Cancel** stops it.
- **Land.** The app has a coarse land mask (coastline from Natural Earth, lenient near the coast) and routes round it, but it knows no depths, shipping lanes or restrictions. A leg that starts or ends close to land is sailed as drawn. Always check the found line on a chart.
- **The result** is an ordinary plan: the line is drawn on the map and everything below (summary, tables, graphs, save, compare, roadbook) works on it. The line lives in the plan. **Keep this line as a route** in the summary adds it to your routes (named "*name* (best)"). **Make again with the latest run** searches again from your route as drawn, with the same settings.

The result is only as good as the forecast and the polar: a front or sharp wind shift can make a different line better than the one found.

**Forecast run.** By default the newest stored run of the model is used. To see how an older forecast would have told it, choose one of the stored runs in the list; it is used wherever it reaches, and the newest run elsewhere. A plan opened later shows the runs it was made with; **Make again with the latest run** in its summary makes a new plan with the same route, departure and model.

The result has two tabs.

- **Summary:** at the top the figures of the passage: which polar was used and its general adjustment, whether the route was sailed, motored or mixed (with the motoring time and share, or "Motor not started"), waiting for wind if it occurred, the forecast models and runs used, departure, ETA, time, route length and distance really sailed (longer when tacking or gybing). Then *Wind along the route* as a list: highest and lowest TWS, the wind speed that was exceeded or not reached 20 % of the time, the average wind direction with its range, and the **maximum gust** (only if the model forecasts gusts). Then *Boat speed*: average SOG, SOG range and average VMG. Below them is the leg table and two histograms.
- **Leg table.** One row for each hour of a leg (a leg shorter than an hour has one row), with a thick line between legs and a last row, **All**, for the whole passage. Each row shows the mean over its time: **Time** (the first column stays in place when you scroll sideways, and the header stays on top when you scroll down), **Dur** (duration), **TWS** (true wind speed), **TWA** (true wind angle to the course), **TWD** (true wind direction), **SOG** (speed over ground), **VMG** (velocity made good along the route), **DTF** (distance to finish in nautical miles at the start of the row) and **FC** (forecast model: H = HARMONIE, E = ECMWF-IFS, A = ARPEGE, with ">" where it changes within the row). TWA is written `60°<` when the wind comes from starboard and `>123°` when it comes from port; a `!` in front means the boat was tacking or gybing for most of the row. `M` replaces the angle when the engine ran most of the row, and `W` when the boat was waiting for wind. Speeds are whole knots for TWS (SOG and VMG have one decimal), or one decimal in m/s, following the unit in Settings. The table is wider than a narrow phone only if the font is large; then it scrolls sideways.
- **Histograms.** The share of the time in each wind speed class (2 kn, or 1 m/s wide) and in each of eight wind directions.
- **Graphs:** wind speed, boat speed (made good along the route solid, over ground dashed), and wind direction, with strips showing which model is used and how the boat is sailed (tacking, gybing, motoring, waiting). On the wind graph the **gust** is a dashed line in the colour of the model that forecast it (HARMONIE and ARPEGE only). Choose the horizontal axis: **Clock**, **Time since departure** or **Distance along the route**. Touch a graph to put a cursor on all of them: the sailboat symbol moves to that point on the map, the map shows the wind of that moment, and the line above the graphs reads the plan there (with the gust, if there is one).

The panel leaves the map visible and usable.

## 7.5 Save, compare, roadbook

- **Save** keeps the plan on the phone with its figures and the settings it was made with. Saved plans are listed in **Routing**. They can be opened again after the forecasts they used have expired.
- Tick two to six saved plans: their lines appear on the map at once, each in its own colour. **Compare ticked plans (2–6)** shows a table of their figures and their graphs on one axis in the same colours, with one boat per plan on the map. Use it to compare departure times or models.
- **Roadbook** saves a single HTML file in the Downloads folder, with the figures (polar, routing, wind, boat speed), the waypoints with times, the same leg table as the summary, the histograms and the graphs, including the gust. It opens in any browser and can be printed.

## 7.6 Best departure

In **Plan a passage...**, tick **Find the best departure: try departures from the time above on**, choose how long (12 to 72 hours) and how often (every 30 to 180 minutes), and **Find**. In the **Avoid** part of the settings tick what you want to avoid:

- **Beating in wind above** a limit (beating means a course within 60° of the wind),
- **Wind above** a limit,
- **Night**.

Each departure gets a score: the trip's hours plus, for each ticked condition, the hours that meet it times its weight. Beating in strong wind therefore counts twice and costs more than strong wind downwind. The best departure has the lowest score; with nothing ticked it is the shortest trip.

The result is a panel under the map, like a plan. The bars show the score of each departure, split into trip time and the ticked conditions (in the colours of the conditions); the best is marked. Touch a bar to read that departure, and **Open this plan** to see and save it. A plan opened from the panel shows on top; closing it brings the panel back.

If **Find the best route** is also ticked, every departure is searched for its best route (a quicker search than a single plan's), the trip time of your route as drawn is shown as a dashed line, and a sentence says what the best route saves. The map draws the routes of the departures: the one you touch thickest, its two neighbours on each side thinner. **Open this plan** then searches that departure again in full.

## 7.7 Parameters

The **Parameters** window (the link beside **Avoid**) holds the general settings of the search:

- **Weights.** Time is the reference: an hour at sea counts as one hour. Three vertical sliders (**Beating**, **Strong wind**, **Night**) say how many hours an hour of that condition adds, from 0 to 3 in tenths; together they can add at most 6, so time always keeps a share. A bar shows the shares and sentences show what an hour of each condition counts as. **Weights to 1** resets them. A weight counts only when its box under **Avoid** is ticked for that plan or sweep. Night is not bad in itself: leave it unticked if you like sailing at night.
- **Integration step.** The isochrone search sails each stage of its search in steps of this length (blank is 10 minutes, 1 to 60). Shorter steps follow the wind more closely and take longer. Leave it blank unless you are trying out how it matters. The grid search has no such setting: its grid is finer on short parts of the route and coarser on long ones, and coarser still in a departure sweep.

## 7.8 What to remember

The forecast is not certain and the plan assumes the boat sails as the polar says, with no current and no waves beyond what the polar and the adjustments describe. Use it to compare alternatives and to see where the hard parts are, not as the only source.

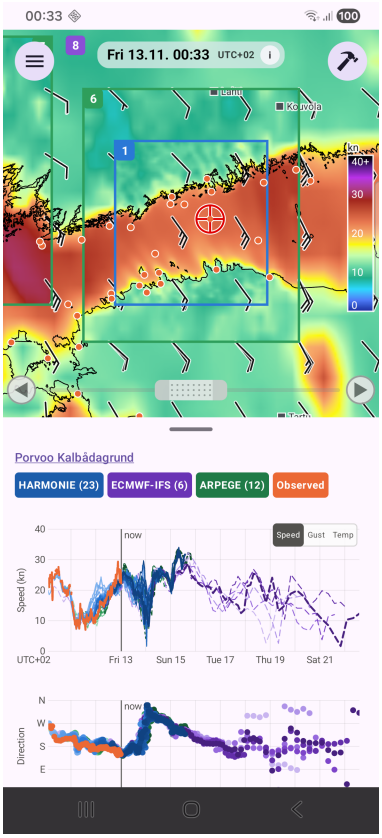



Figure 6: Station

# 8 Stations

The small orange dots on the map are marine observation stations. Tap one to see its **measured wind** (and, where the station has a thermometer, its **air temperature** on the Temp chart) drawn on the charts together with the forecasts, so you can see how well recent runs matched reality.

## 8.1 Sources

| Source | Data |
|---|---|
| FMI | 10-minute observations |
| Marine Helsinki | Kruunuvuorenselkä |
| SMHI (Sweden) | hourly |
| DMI (Denmark) | 10-minute; gust is the highest 3-second mean |
| TalTech | Tallinnamadal, latest values only |
| Estonian Environment Agency | 12 coastal stations: 10-minute history of the last few days (the agency keeps about 3.7 days); the other stations: current wind only, as a text label |

Tallinnamadal and the Estonian stations that are not among the 12 coastal ones show only the wind of the moment as a label on the map, not a chart, because those sources offer no history. The chart never reaches further back than the oldest forecast run at the tapped point. Where a source has no gust or direction, those lines are left out. A gap in the line means the station was silent.

Station dots are 2 dp in a wide view and grow to 12 dp from zoom level 10.

# 9 Settings

All settings are in the hamburger menu.

| Setting | Meaning |
|---|---|
| **Add new forecast tile**, tile rows | See First start and Tiles |
| **Update all active areas** | Starts an update now (shown once there is a tile) |
| **Routing** | Routes, passage plans and saved plans; see Routing |
| **Show the no-go zone of weather routing** | Tick to draw the zone (land and the coast next to it) that weather routing will not cross; see Routing |
| **Export GRIB2 of active tiles to Downloads** | Copies the newest raw GRIB2 of every active tile to Downloads, with the tile number in the name |
| **kn / m/s** | Speed unit, default knots. Barbs are always in knots |
| **Wind colours**, **Wind barbs** | Tick each layer on or off |
| **Grey land** | Untick to show only the shoreline |
| **UI Color theme** | Automatic, light or dark; tap to cycle. The offline manual follows it |
| **Data limit** | 1–5 GB (default 2) |
| **History kept** | 1–14 days of runs to keep, set with a slider (applied when you close the dialog) |
| **ARPEGE key** | Your Météo-France key, needed for ARPEGE; see ARPEGE |
| **Update log...** | See Updates |
| **Manual** | This manual, stored in the app, so it works without a connection |
| **About WindBrowser** | Credits, version, and links to the online manual and the PDF |
| **Clear all tiles** | Removes everything; always asks first |

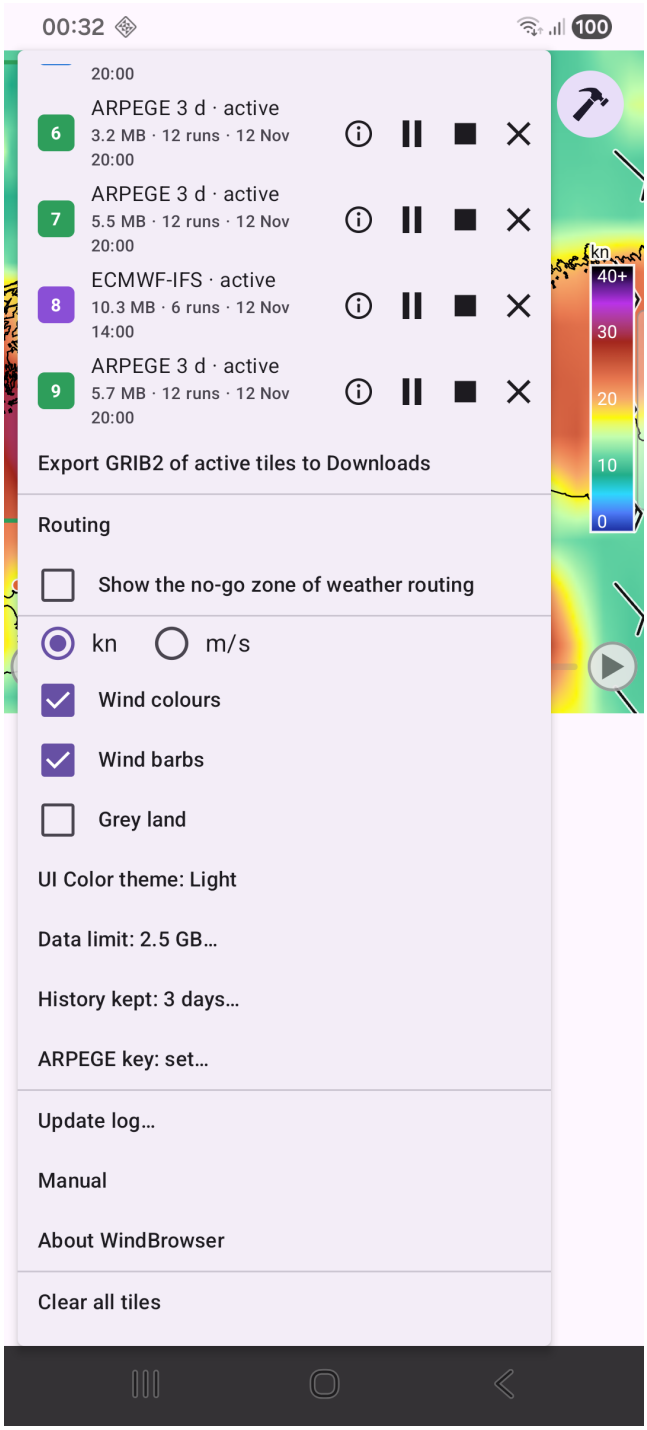



Figure 7: Menu

The boat and routing settings are not in the menu: they belong to each plan and are in **Plan a passage…** (Routing).

### 9.1 Backup

The app takes part in Android's backup, so a new phone restored from your Google backup gets your settings, areas (the tile list), routes, polars, saved plans and routing settings back. The downloaded forecast runs and the update log are not backed up; they are fetched again, but the history starts over.

## 10 Updates and troubleshooting

### 10.1 How updates work

- A background job runs about **hourly** and updates every active tile.
- The app also updates when you bring it to the front, at most every 10 minutes.
- Because Android may put the hourly job to sleep while the phone idles, an **alarm every hour** also asks for an update.
- FMI keeps **only its latest run**, and the app takes only the newest run of ARPEGE too. A run missed because the phone was idle cannot be fetched later, so open the app now and then in the days before you need it.
- A new ARPEGE run takes about three minutes to download, because Météo-France sends it in many small requests (ARPEGE).

### 10.2 Battery use: set the app to Unrestricted

Android saves power by delaying background work, most of all overnight, and a forecast run that appears meanwhile is lost for good, because FMI keeps only the latest one. Letting WindBrowser run without battery restrictions fixed this in practice: the updates arrive on time and the forecast history has no gaps.

The app asks about this **once**, right after you add your first area. **Allow** shows Android's own question ("Let app always run in background?"). The app checks for new forecasts only about once an hour and downloads only when there is a new run, so the extra battery use is small. If you chose **Not now**, or want to undo it, change it any time in Android's settings: **App info → Battery → Unrestricted** (the wording differs between phone makers). The update log tells whether the app is unrestricted at the moment.

### 10.3 Update log

Menu → **Update log…** lists each job (hourly, alarm or manual, attempt, standby bucket, battery unrestricted or optimised, dozing or not), the result for each tile, and the end. Use it to find out why a run is missing.

### 10.4 Troubleshooting

| Problem | What to try |
|---|---|
| "Downloading the first forecast…" does not finish | Check the connection; see the update log |
| Tile is grey | Its newest run is older than 24 h. Check that it is active and that the phone is online |
| "No forecast at the tapped point" | The point is outside every tile. Add an area (ECMWF-IFS works anywhere) |
| Runs disappeared | The data limit or history limit pruned them; raise it |
| HARMONIE area refused | Larger than 16 square degrees, or fully covered by existing tiles |
| An ARPEGE tile says it needs a key, or "key was rejected" | Enter a key, or a new one if it expired or was replaced: ARPEGE |
| Android delays updates, or runs are missing overnight | Set the app's battery use to Unrestricted (see above); opening the app also triggers an update |

# 11 About and credits

WindBrowser keeps forecast runs for comparison. The version is shown in the menu under **About WindBrowser**.

## 11.1 Where to find the manual

- **In the app, offline:** menu → **Manual**.
- **Online (latest):** https://pclusius.github.io/windbrowser-docs/
- **PDF:** https://pclusius.github.io/windbrowser-docs/WindBrowser-manual.pdf, also linked from **About WindBrowser**.
- **PDF in Finnish (suomeksi):** https://pclusius.github.io/windbrowser-docs/WindBrowser-manual-fi.pdf. It is updated only after major changes, so the English manual is the most current.

## 11.2 Data and credits

- Forecasts: **Finnish Meteorological Institute** open data (HARMONIE and ECMWF IFS), and **Météo-France** (ARPEGE; Open Licence 2.0, through the Météo-France API with your own key).
- Observations: FMI, SMHI, DMI, Marine Helsinki, TalTech and the Estonian Environment Agency (ilmateenistus.ee). Licences are listed in the app.
- Coastline: OpenStreetMap contributors (land polygons) and Natural Earth.
- Place names: GeoNames (CC BY 4.0).
- Weather routing: the isochrone method of Y. Chen and W. Mao, "An Isochrone-Based Predictive Optimization for Efficient Ship Voyage Planning and Execution", *IEEE Transactions on Intelligent Transportation Systems* 25 (11), 18078–18092, 2024, doi:10.1109/TITS.2024.3416349, adapted for sailing boats.

No account is needed. Your position, if you use the location button, is never stored or sent.